

Comparison of some computer vision boards

	Arduino Nicla Vision	OpenMV Cam H7	HuskyLens	VC Picosmart	iVS-AWV3-AR0521
Processor	STM32H757AI16 Dual Arm Cortex M7/M4	ARM 32-bit Cortex-M7 CPU	Kendryte K21 (AI Chip with 64bit RISC-V dual core CPU)	FPGA module, High-end FPU processor	Allwinner V3 system on module
Camera/ image sensor	GC2145 (2MP colour camera)	OV7725 image sensor (can be interfaced with other sensors too)	HuskyLens – OV2640 (2MP camera), HuskyLens PRO – OV5640 (5MP camera)	1-Megapixel CMOS sensor	ON Semiconductor AR0521 sensor module
Notable characteristics and features	MP34DT06JTR digital MEMS microphone, LSM6DSOXTR 6-axis IMU, NXP SE050C2 Crypto chip	Double-precision floating point unit (FPU), Hardware JPEG encoding	It can be used as a stand-alone product or as a peripheral for a microcontroller or microcomputer	VC realtime-OS for real-time image processing	H.264, 1080p60 video codec, integrated ISP supports up to 8MP with auto image function
Connectivity and communication	Bluetooth, Wi-Fi, SPI, I2C, UART	Wi-Fi, BLE, SPI, I2C, UART, CAN	UART, I2C	FPC connector	Multiple wireless connectivity options available
Supported battery/ power source	Li-ion/LiPo single cell, 3.7V	3.7V LiPo battery	Two power pins on the 4-pin JST connector, micro USB connector	N/A	N/A
Suitable for	Asset tracking, object recognition, predictive maintenance	Object detection, line detection, video recording, QR code detection/ decoding, eye tracking, colour tracking, frame differencing	Face recognition, object tracking, object recognition, line tracking, colour recognition, tag recognition	Object recognition, position control, barcode reading, Web edge and level control	Autonomous vehicle and aftermarket home automation camera systems
Compatibility/ integrations	Arduino IDE, Arduino CLI, Arduino IoT Cloud, OpenMV IDE	Edge impulse CLI, OpenMV IDE.	Mind+Editor, Arduino IDE	N/A	N/A
Primary programming language	C/C++	Python	C/C++	N/A	N/A
Edge capability	Yes	Yes	Yes	Yes	Yes
Industry-ready?	Yes	Yes	More suited to prototyping	Yes	Yes
Dimensions in mm (l x w x h)	22.86 x 22.86	45 x 36 x 30	52 x 44.5	22 x 23.5	N/A
Weight	2gm	19gm	N/A	N/A	N/A
MSRP	\$115.00	\$65.00	\$54.90	Quote based	Quote based
Datasheet	https://docs.arduino.cc/static/592480018567f53345d4f2b18199cd67/ABX00051-datasheet.pdf	https://cdn.shopify.com/s/files/1/0803/9211/files/OpenMV-H7.pdf?10726770933929369108	https://wiki.dfrobot.com/HUSKYLENS_V1.0_SKU_SEN0305_SEN0336	https://www.vision-components.com/en/products/oem/arm-linux/board-cameras/vc-picosmart/	https://www.ienso.com/product/ivs-awv3-ar0521/